

CBSE 2027 Maths — Predicted Paper (Set 5 of 10)

Based on the highly weighted CBSE syllabus chapters and the official competency-based blueprint (with a focus on analytical thinking, calculus integration, and spatial geometry), this predicted paper (Set 5) features all-new, high-probability scenarios.

SECTION A: Objective Type Questions (1 Mark Each x 20 = 20 Marks)

Q1. Let $A = \{1, 2, 3\}$. How many relations can be defined on set A that contain the elements $(1, 2)$ and $(1, 3)$, and are reflexive and symmetric but not transitive? * **Chapter:** Relations and Functions * **Probability Tag: Medium**

Q2. Find the principal value of $\tan^{-1}(\sqrt{3}) - \sec^{-1}(-2)$. * **Chapter:** Inverse Trigonometric Functions * **Probability Tag: Very High**

Q3. If a matrix $A = \begin{pmatrix} \alpha & \beta \\ \gamma & -\alpha \end{pmatrix}$ is such that $A^2 = I$ (where I is the identity matrix), which of the following relationships is correct? (a) $1 + \alpha^2 + \beta\gamma = 0$ (b) $1 - \alpha^2 + \beta\gamma = 0$ (c) $1 - \alpha^2 - \beta\gamma = 0$ (d) $1 + \alpha^2 - \beta\gamma = 0$ * **Chapter:** Matrices * **Probability Tag: High**

Q4. Evaluate the determinant: $\Delta = \begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix}$. * **Chapter:** Determinants * **Probability Tag: Very High**

Tag: Very High

Q5. What is the derivative of $\sin^2 x$ with respect to $\cos^2 x$? * **Chapter:** Continuity and Differentiability * **Probability Tag: High**

Q6. The tangent to the curve $y = e^{2x}$ at the point $(0, 1)$ meets the x-axis at which point? * **Chapter:** Application of Derivatives * **Probability Tag: Medium**

Q7. Evaluate the indefinite integral: $\int \frac{dx}{x^2+16}$. * **Chapter:** Integrals * **Probability Tag: Very High**

Q8. Find the area bounded by the curve $y = |x|$, the x-axis, and the ordinates $x = -1$ and $x = 1$. * **Chapter:** Application of Integrals * **Probability Tag: High**

Q9. Write the order of the differential equation: $\frac{d^2y}{dx^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$. * **Chapter:** Differential Equations * **Probability Tag: Very High**

Q10. If $\vec{a}$ and $\vec{b}$ are two unit vectors such that $|\vec{a} + \vec{b}| = \sqrt{3}$, find the angle between them. * **Chapter:** Vector Algebra * **Probability Tag: Very High**

Q11. Find the value of λ for which the vectors $2\hat{i} - \lambda\hat{j} + \hat{k}$ and $\hat{i} + 2\hat{j} - \hat{k}$ are orthogonal (perpendicular). * **Chapter:** Vector Algebra * **Probability Tag: High**

Q12. What is the distance of the point $(3, 4, 5)$ from the origin? * **Chapter:** Three-Dimensional Geometry * **Probability Tag: Very High**

Q13. Write the direction ratios of the normal to the plane $2x - y + 2z = 5$. * **Chapter:** Three-Dimensional Geometry * **Probability Tag: Medium**

Q14. In a Linear Programming Problem, the objective function is $Z = px + qy$ ($p, q > 0$). If the maximum value of Z occurs at both the corner points $(3, 4)$ and $(4, 3)$, what is the relation between p and q ? * **Chapter:** Linear Programming * **Probability Tag: High**

Q15. If $P(A) = 0.5$ and $P(B) = 0$, then what is the value of $P(A|B)$? * **Chapter:** Probability * **Probability Tag: Very High**

Q16. Evaluate the definite integral: $\int_{-\pi}^{\pi} x^{10} \sin^7 x \, dx$. * **Chapter:** Integrals * **Probability Tag: Very High**

Q17. Find the integrating factor of the differential equation: $x \frac{dy}{dx} - y = 2x^3$. * **Chapter:** Differential Equations * **Probability Tag: High**

Q18. A card is drawn at random from a standard deck of 52 playing cards. What is the probability of drawing a non-face card? * **Chapter:** Probability * **Probability Tag: Medium**

Q19. (Assertion-Reason): Assertion (A): The vectors $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ and $\vec{b} = -4\hat{i} + 6\hat{j} - 8\hat{k}$ are collinear. **Reason (R):** Two vectors are collinear if one is a scalar multiple of the other. * **Chapter:** Vector Algebra * **Probability Tag: Very High**

Q20. (Assertion-Reason): Assertion (A): If $|A| = 0$ for a square matrix A , then A is invertible. **Reason (R):** A square matrix A is invertible if and only if it is non-singular ($|A| \neq 0$). * **Chapter:** Determinants * **Probability Tag: High**

SECTION B: Very Short Answer Type Questions (2 Marks Each x 5 = 10 Marks)

Q21. Solve for x : $\sin^{-1}(1 - x) - 2 \sin^{-1} x = \frac{\pi}{2}$. * **Chapter:** Inverse Trigonometric Functions * **Probability Tag: High**

Q22. Find $\frac{dy}{dx}$ if $x = at^2$ and $y = 2at$ (where a is a constant). * **Chapter:** Continuity and Differentiability * **Probability Tag: Very High**

Q23. Find a unit vector perpendicular to both the vectors $\vec{a} = 2\hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + 2\hat{k}$. * **Chapter:** Vector Algebra * **Probability Tag: Very High**

Q24. Solve the differential equation: $\frac{dy}{dx} = 1 + x + y + xy$. * **Chapter:** Differential Equations * **Probability Tag: High**

Q25. A random variable X has a probability distribution $P(X = x) = kx$ for $x = 1, 2, 3$. Find the value of constant k and calculate $P(X \geq 2)$. * **Chapter:** Probability * **Probability Tag: High**

SECTION C: Short Answer Type Questions (3 Marks Each x 6 = 18 Marks)

Q26. Let R be the relation defined on the set of real numbers $\mathbb{R}$ as $R = \{(a, b) : a - b \text{ is an integer}\}$. Prove that R is an equivalence relation. * **Chapter:**

Relations and Functions * **Probability Tag: High**

Q27. If $A = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$, show that $A^2 - 5I = O$. Hence, find A^{-1} . * **Chapter:** Matrices *

Probability Tag: Very High

Q28. Differentiate $\tan^{-1} \left(\frac{3x-x^3}{1-3x^2} \right)$ with respect to $\sec^{-1} \left(\frac{1}{2x^2-1} \right)$, for $0 < x < \frac{1}{\sqrt{3}}$. *

Chapter: Continuity and Differentiability * **Probability Tag: Very High**

Q29. Evaluate the definite integral using standard properties: $\int_0^{\pi/2} \frac{\sin x - \cos x}{1 + \sin x \cos x} dx$. * **Chapter:** Integrals * **Probability Tag: High**

Q30. Find the particular solution of the differential equation $(x^2 - y^2)dx + 2xy dy = 0$, given that $y = 1$ when $x = 1$. * **Chapter:** Differential Equations * **Probability Tag: Very High**

Q31. Two dice are thrown simultaneously. Find the probability of getting an even number on the first die or a total sum of 8 on both dice. * **Chapter:** Probability * **Probability Tag: Medium**

SECTION D: Long Answer Type Questions (5 Marks Each x 4 = 20 Marks)

Q32. Evaluate the integral: $\int_0^{\pi} \frac{x \tan x}{\sec x + \tan x} dx$. * **Chapter:** Integrals * **Probability Tag: Very High**

Q33. Prove that the volume of the largest cone that can be inscribed in a given sphere of radius R is $\frac{8}{27}$ of the volume of the sphere. * **Chapter:** Application of Derivatives * **Probability Tag: Very High**

Q34. Find the distance of the point $(2, 12, 5)$ from the point of intersection of the line $\vec{r} = 2\hat{i} - 4\hat{j} + 2\hat{k} + \lambda(3\hat{i} + 4\hat{j} + 2\hat{k})$ and the plane $\vec{r} \cdot (\hat{i} - 2\hat{j} + \hat{k}) = 0$. * **Chapter:** Three-Dimensional Geometry * **Probability Tag: Very High**

Q35. Using the method of integration, find the area of the region bounded by the circle $x^2 + y^2 = 4$ and the line $x + y = 2$ in the first quadrant. * **Chapter:** Application of Integrals * **Probability Tag: High**

SECTION E: Case-Study Based Questions (4 Marks Each x 3 = 12 Marks)

Q36. Case Study 1: Vector Algebra (Drone Spraying) A farmer uses three agricultural drones to spray fertilizer on his triangular field. The drones are stationed at three points in the field representing the vertices of the triangle, defined by the coordinates $A(1, 1, 1)$, $B(2, 3, 5)$, and $C(1, 5, 5)$. (i) Find the position vector $\vec{AB}$ and its magnitude. (1 Mark) (ii) Find the unit vector perpendicular to the plane containing the drones (i.e., plane ABC). (1 Mark) (iii) Using vector methods, calculate the total area of the triangular field ABC covered by the drones. (2 Marks) * **Chapter:** Vector Algebra * **Probability Tag: Very High**

Q37. Case Study 2: Matrices (School Awards) Three schools A, B, and C decided to organize a fair for collecting funds to award their students for three core values: Honesty, Punctuality, and Discipline. School A awards ₹ x , ₹ y , and ₹ z to 3, 2, and 1 students respectively, distributing a total of ₹1000. School B awards 4, 1, and 3 students respectively for these values, distributing a total of ₹1500. School C awards 2, 3, and 1 students respectively for these values, distributing a total of ₹1200. (i) Represent the above situation as a system of linear equations. (1 Mark) (ii) Express the system as a matrix equation of the form $AX = B$. (1 Mark) (iii) Using the matrix inverse method (or determinants), find the exact award amount for Honesty (x) and Punctuality (y). (2 Marks) * **Chapter:** Matrices and Determinants *

Probability Tag: Very High

Q38. Case Study 3: Probability (Exam Guessing) In a competitive examination, a student faces a section of 10 multiple-choice questions. For each question, there are 4 options out of which exactly 1 is correct. A student who has not studied at all decides to randomly guess the answer to every single question. (i) What is the probability of a success (guessing correctly) and a failure (guessing incorrectly) for any single question? (1 Mark) (ii) Using the binomial distribution, write the generalized probability function $P(X = x)$ for getting exactly x correct answers out of 10. (1 Mark) (iii) What is the expected number of correct answers (Mean) the student will get? Also, find the variance of this binomial distribution. (2 Marks) *

Chapter: Probability * **Probability Tag: High**

CBSE Class 12 (2027) — AI-Predicted via PYQ/Blueprint Analysis. Probability-weighted guidance, not actual exam questions.